

Seat No:

SET P

AUTONOMOUS STATE EXAMINATION BOARD**Btech Computer Scie Engg DBMS****Subject: Data Structure · Pattern: CBCS Pattern****Day & Date:** Thursday, September 17, 2026 **Duration:** 180 Minutes **Max. Marks:** 70 Marks**Instructions for Candidates:**

1. Q.1 is compulsory. Mention the question paper set (**P**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. Standard University Marking Scheme

Q.1 Choose the correct alternatives for the following questions. [14 Marks]

1. 14) Which of the following is not a cor Partitioning a) c) Buddy system ntiguous memory management technique? b) Paging d) Over lays
 - a) A
 - b) C
 - c) Paging
 - d) Over lays
2. 158. Identify the incorrect pair. (1) Lectins - Concanavalin A (2) Drugs - Ricin (3) Alkaloids - Codeine (4) Toxin - Abrin
 - a) Option A
 - b) Option B
 - c) Option C
 - d) Option D
3. Two bodies A and B of same mass undergo 6 completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2 . The ratio $v_1 : v_2$ is : (2) 2:1 (1) 1:2 (3) 4:1 (4) 1:4 In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is $9.8 \times 10^6 \text{ kg m}^2$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5} \text{ Am}^2$; then the value of 'x' is : (1) $5\pi^2$ (2) $128 \pi^2$ (3) $50 \pi^2$ (4) $1280 \pi^2$ [Contd...
 - a) 2:1
 - b) 1:2
 - c) 4:1
 - d) 1:4

4. 160. Which of the following is not an objective of Biofortification in crops ? (1) Improve vitamin content (2) Improve micronutrient and mineral content (3) Improve protein content (4) Improve resistance to diseases
- Option A
 - Option B
 - Option C
 - Option D
5. a) Explain the Round Robin (RR) Scheduling algorithm with an example. Find the average waiting time and average turnaround time for RR scheduling algorithm for the following processes with a time quantum of 4 ms. Process CPU Burst Time P1 10 P2 5 P3 2 P4 3 b) Write short note on:
- Write short note on:
6. 37. A series LCR circuit containing 5.0 H inductor,
- Option A (Inspect Question Image)
 - Option B (Inspect Question Image)
 - Option C (Inspect Question Image)
 - Option D (Inspect Question Image)
7. 28 [Contd...
- Option A (Inspect Question Image)
 - Option B (Inspect Question Image)
 - Option C (Inspect Question Image)
 - Option D (Inspect Question Image)
8. 7) The network IP address or logical address present for each node in the network is of bits. a) 12 b) 32 c) 64 d) 48
- 12
 - 32
 - 64
 - 48
9. 34. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.
- Option A (Inspect Question Image)
 - Option B (Inspect Question Image)
 - Option C (Inspect Question Image)
 - Option D (Inspect Question Image)
10. 23 [Contd...
- Option A (Inspect Question Image)
 - Option B (Inspect Question Image)

- c) Option C (Inspect Question Image)
- d) Option D (Inspect Question Image)

- 10. Option A (Inspect Question Image)
- 11. Option B (Inspect Question Image)
- 12. Option C (Inspect Question Image)
- 13. Option D (Inspect Question Image)

Option A (Inspect Question Image)
Option B (Inspect Question Image)
Option C (Inspect Question Image)
Option D (Inspect Question Image)

The moment of inertia of a thin rod about an axis 14 passing through its mid point and perpendicular to the rod is 2400 g cm^2 . The length of the 400 g rod is nearly : (2) 17.5 cm (1) 8.5 cm (3) 20.7 cm (4) 72.0 cm

- a) 17.5 cm
- b) 8.5 cm
- c) 20.7 cm
- d) 72.0 cm

What is the fundamental law governing thermodynamics process #30 in Chemistry?

- a) Option Alpha for Q-PAPER-SRC-NEET-A-30
- b) Option Beta (Correct) for Q-PAPER-SRC-NEET-A-30
- c) Option Gamma for Q-PAPER-SRC-NEET-A-30
- d) Option Delta for Q-PAPER-SRC-NEET-A-30

— END OF QUESTION PAPER (P) —